Physics Galaxy
Telegram Test Series

JEE Main Level Practice Test-6
for
JEE & NEET Aspirants

Topic: EXPANSION, CALORIMETRY, KTG & THERMODYNAMICS **Time: 75Min Marking: +4 –1**

Section - A : MCQs with Single Option Correct

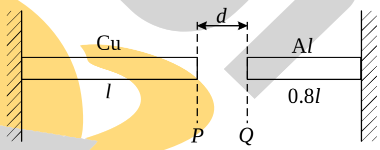
1. A pendulum clock keeps correct time at 20°C. The correction to be made during summer per day, where the average temperature is 40°C ($\alpha = 10^{-5}$/°C) will be :

(A) 5.64 sec (B) 6.64 sec (C) 7.64 sec (D) 8.64 sec

2. When the temperature of water rises, then its density will :

(A) Increase (B) Decrease
(C) May increase or decrease (D) Remain same

3. The copper bar and the aluminium bar in the figure are attached to an immovable wall, the length of bars are l and $0.8\,l$ respectively. At $\theta°C$, the gap between rods is $d(d << l)$. At what temperature, the ends P and Q just touch to each other : ($\alpha_{Cu} = \alpha_0$, $\alpha_{Al} = 1.2\alpha_0$)



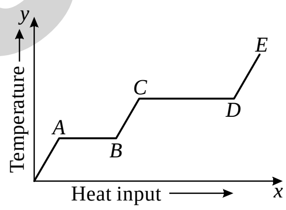
(A) $\theta_0 + \dfrac{1.51d}{l\alpha_0}$ (B) $\theta_0 + \dfrac{1.2d}{l\alpha_0}$

(C) $\theta_0 + \dfrac{0.51d}{l\alpha_0}$ (D) $\theta_0 + \dfrac{d}{l\alpha_0}$

4. Two identical glass spheres filled with air are connected by a horizontal glass tube. The glass tube contains a pellet of mercury at its mid point. Air in one sphere is at $0°C$ and the other is at 20°C. If both the vessels are heated through 10°C then neglecting the expansions of the bulbs and the tube. Ignore the conduction of heat by the mercury pellet.

(A) The mercury pellet gets displaced towards the sphere at lower temperature.
(B) The mercury pellet gets displaced towards the sphere at higher temperature.
(C) The mercury pellet does not get displaced at all.
(D) The temperature rise causes the pellet to expand without any displacement.

5. A solid substance is supplied heat at a constant rate and the variation of temperature with heat input is shown in the figure. Choose the correct statement :



(A) The latent heat of vaporization is half the latent heat of fusion
(B) The specific heat in the solid state is less than that in the liquid state
(C) The specific heat in the solid state is less than that in the liquid state
(D) AB and CD represent changes of phase from liquid to vapour and from solid to liquid respectively

6. The coefficient of apparent expansion of a liquid is C when heated in a copper vessel and it is S when heated in silver vessel. If A is the coefficient of linear expansion of copper, then that of silver is :

(A) $\dfrac{C+S-3A}{3}$ (B) $\dfrac{C+3A-S}{3}$ (C) $\dfrac{S+3A-C}{3}$ (D) $\dfrac{C+S+3A}{3}$

7. The specific heat of the same substance is expressed in two units; C_1 cal/gm°C and C_2 cal/gm°F. Then which of the following relation is true?

(A) $C_1 > C_2$ (B) $C_1 = C_2$

(C) $C_1 < C_2$ (D) C_1 & C_2 cannot be compared

8. A metal rod of Young's modulus Y and coefficient of thermal expansion α is held at its two ends such that its length remains invariant. If its temperature is raised by t°C, the linear stress developed in it is:

(A) $\frac{\alpha t}{Y}$ (B) $\frac{Y}{\alpha t}$ (C) $Y\alpha t$ (D) $\frac{1}{(Y\alpha t)}$

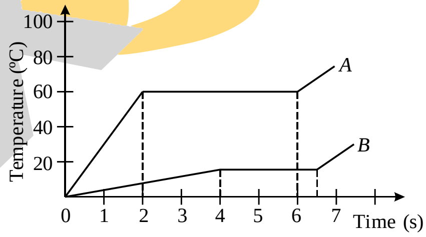
9. A block of ice with a lead shot embedded in it is floating on water container in a vessel. The temperature of the system is maintained at 0°C as the ice melts. When the ice melts completely, the level of water in the vessel :

(A) rises (B) falls (C) remains same (D) none of these

10. Two marks on a glass rod 10 cm apart are found to increase their distance by 0.08 mm when the rod is heated from 0°C to 100°C. A flask made of the same glass as that of rod measures a volume of 1000 cc at 0°C. The volume it measures at 100°C (in cc) is:

(A) 1002.4 (B) 1004.2 (C) 1006.4 (D) 1008.2

11. Two substances A and B of equal mass m are heated by uniform rate of 6 cal s^{-1} under similar conditions. A graph between temperature and time is shown in figure. Ratio of heat absorbed H_A/H_B by them for complete fusion is :



(A) $\frac{9}{4}$ (B) $\frac{4}{9}$

(C) $\frac{8}{5}$ (D) $\frac{5}{8}$

12. Two different masses m and $2m$ of same ideal gas are heated separately in vessels of equal volume. T-P curve for mass $2m$ makes angle α with T-axis and that for mass m makes angle β with T-axis then :

(A) $\tan \alpha = \tan \beta$ (B) $\tan \alpha = 2 \tan \beta$ (C) $\tan \beta = 2\tan \alpha$ (D) None of these

13. A fixed mass of gas is trapped in a syringe. The volume of the syringe is slowly reduced, compressing the gas without changing the temperature. The pressure exerted on the walls of the syringe changes because :

(A) The particles have more kinetic energy

(B) The particles collide with each other more often

(C) The particles hit the walls of the syringe with more force

(D) The particles hit the walls of the syringe more often

14. Two containers of equal volume contain the same gas at pressure p_1 and p_2 and absolute temperatures T_1 and T_2 respectively. On joining the vessels, the gas reaches a common pressure p and a common temperature T. The ratio p/T is equal to :

(A) $\frac{p_1}{T_1} + \frac{p_2}{T_2}$ (B) $\frac{1}{2}\left[\frac{p_1}{T_1} + \frac{p_2}{T_2}\right]$ (C) $\frac{p_1 T_2 + p_2 T_1}{T_1 + T_2}$ (D) $\frac{p_1 T_2 - p_2 T_1}{T_1 - T_2}$

15. A meter washer has a hole of diameter d_1 and external diameter d_2, where $d_2 = 3d_1$. On heating, d_2 increases by 0.3%. Then d_1 will :

(A) decrease by 0.1% (B) decrease by 0.3% (C) increase by 0.1% (D) increase by 0.3%.

16. An ideal gas filled in a cylinder occupies volume V. The gas is compressed isothermally to the volume $V/3$. Now the cylinder valve is opened and the gas is allowed to leak keeping temperature same. What percentage of the number of molecules should escape to bring the pressure in the cylinder back to its original value?

(A) 66% (B) 33% (C) 0.33% (D) 0.66%

17. A and B are two identical containers. A contains 5 moles of H_2 and B contains 10 moles of O_2 at same temperature. Assuming real gas behaviour, match the following columns :

Column - I	**Column - II**
(i) In container A	(p) Pressure is more
(ii) In container B	(q) RMS speed is more
(iii) As, number of moles is more	(r) Average thermal energies are same
(iv) Due to a smaller molecular mass	(s) Internal energy is more

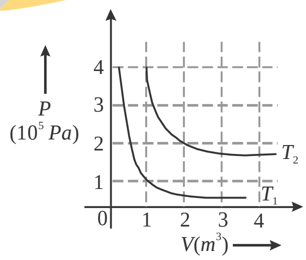
Code :

	i	**ii**	**iii**	**iv**
(A)	q, r	p, r, s	p, s	q
(B)	q	p	r	s
(C)	q, r	p, r	s	q
(D)	r	p	s	q

18. As an air bubble rises from the bottom of a lake to the surface such that its temperature is constant and volume is doubled. If the atmospheric pressure is 76.0 cm of Hg and the density of mercury is 13.6 g/cc, the depth of the lake is : (neglect the surface tension of air bubble).

(A) 5.17 m (B) 20.6 m (C) 7.60 m (D) 10.3 m

19. The following graph shows two isotherms for a fixed mass of an ideal gas. The ratio of r.m.s. speed of the molecules at temperature T_1 and T_2 is :



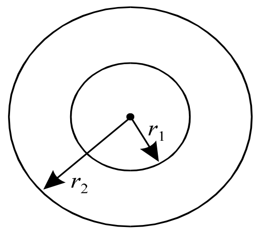
(A) $2\sqrt{2}$

(B) $\sqrt{2}$

(C) 1/2

(D) 1/4

20 If an annular disc of radii r_1 and r_2 is heated, then :



(A) r_1 increases, r_2 decreases

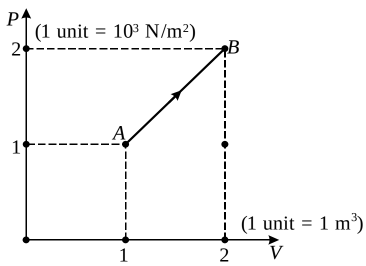
(B) r_2 increases, r_1 decreases

(C) both r_1 and r_2 increase

(D) r_2 increases, r_1 remains unchanged

<u>Section- B: INTEGER Answer Type Questions</u>

21. For P-V diagram shown for a diatomic ideal gas, calculate the heat supplied to the gas from state A to B (in kJ).

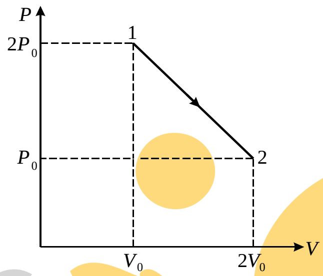


22. If the P-V diagram of a diatomic gas is plotted, it is a straight line passing through the origin. The molar heat capacity of the gas in this process is IR, where I is an integer and R is universal gas constant. Find the value of I.

23. A simple seconds pendulum is constructed out of a very thin string of thermal coefficient of linear expansion $\alpha = 20 \times 10^{-4}$ /°C and a heavy particle attached to one end. The free end of the string is suspended from the ceiling of an elevator at rest. The pendulum keeps correct time at 0°C. When the temperature rises to 50°C, the elevator operator of mass 60 kg being a student of Physics accelerates the elevator vertically, to have the pendulum correct time. Calculate the apparent weight (N) of the operator when the pendulum keeps correct time at 50°C (Take $g = 10$ m/s^2)

24. Two identical conducting rods are first connected independently (in parallel) to two vessels, one containing water at 100°C and the other containing ice at 0°C. In the second case, rods are joined end to end and are connected to the same vessels. If q_1 and q_2 (in g/s) are the rates of melting of ice in two cases, then find the ratio of q_1/q_2.

25. An ideal gas undergoes a process shown in the P-V diagram. If the change in heat content of the gas in this process is $\frac{K}{2} P_0V_0$. Then find the value of K.



26. One mole of an ideal gas undergoes a thermodynamic process whose molar specific heat is given by the equation $C = C_V + 4R$. Equation of such a process is given by VT^{-n}, calculate n.

27. Steam at 100°C is added slowly to 1400 gm of water at 16°C until the temperature of water is raised to 80°C. Calculate the mass (in gm) of steam required to do this. ($L_v = 540$ cal/gm)

28. One gram of ice at 0°C is added to 5 gram of water at 10°C. If the latent heat is 80 cal, calculate the final temperature (°C) of the mixture. [$S_W = 1$ cal/gm°C, $L_{ice} = 80$ cal/gm]

29. Two identical containers contain same amount of ideal gas at identical condition. Gas in one container is expanded to 4 times of its volume at constant pressure and gas in another container is expanded to four times of its volume by isothermal process. If W_1 and W_2 be the respective work done, then $\frac{W_1}{W_2} = \frac{\alpha}{\ln 2}$. Find $\frac{2}{3}\alpha$.

30. One mole of ideal gas whose adiabatic exponent $\gamma = \frac{4}{3}$ undergoes a thermodynamics process $P = 200 + \frac{1}{V}$. Calculate the change in internal energy of gas when volume changes from $2m^3$ to $4m^3$.

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Section - A : MCQs with Single Option Correct

1. (D)	**2.** (C)	**3.** (C)	**4.** (B)
5. (C)	**6.** (B)	**7.** (A)	**8.** (C)
9. (B)	**10.** (A)	**11.** (C)	**12.** (B)
13. (D)	**14.** (B)	**15.** (D)	**16.** (A)
17. (A)	**18.** (D)	**19.** (C)	**20** (C)

Section- B: INTEGER Answer Type Questions

21. [9]	**22.** [3]	**23.** [660]	**24.** [4]
25. [3]	**26.** [4]	**27.** [160]	**28.** [0]
29. [1]	**30.** [1200]		